Rotation periods and spin senses of asteroids from Gaia DR3 sparse photometry: calibrated reliability, null tests and disputed LCDB periods

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Abstract

I derive rotation periods and spin senses for numbered asteroids with at least 25 Gaia DR3 transits, giving 41,842 solutions. The model fits a separate brightness level for each observing season, a linear phase term and a second-order Fourier series. The rotational phase is corrected for the phase-angle bisector, with both senses of rotation tried. Known LCDB periods are split once into a tuning half and a test half: the period-doubling rule and a logistic model of the probability of a correct period are fitted on the first, and the confidence threshold comes from null tests that preserve the Gaia sampling but remove rotation. 6,238 periods reach the confident tier. On the test half, 89.3% of them are correct within 3%, and 95.4% where the reference quality is $U \geq 3-$; injected synthetic lightcurves are recovered with 96–99% precision. The spin sense agrees with published poles for 98.2% of asteroids in an analysis of the same data and for 94.4% of independent DAMIT models. The senses reproduce the known Yarkovsky segregation in collisional families: 74.8% of 2,345 members lie on the predicted side, compared with 52.4 ± 0.9% when senses are permuted within families, and the fraction rises to 82% for the most drifted members. I identify 108 LCDB periods disputed by independent survey data or by agreement among Gaia DR3 analyses. Nine more were already revised in the 2023 LCDB to the values found here. I list 861 candidate new periods, with at most 4.5% expected noise.

Keywords: Asteroids, rotation, Asteroids, dynamics, Photometry, Data reduction techniques

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1. Introduction

Gaia (Gaia Collaboration et al., 2016) released sparse photometry for about 157,000 numbered asteroids in DR3 (Gaia Collaboration et al., 2023a,b). Sparse photometry can constrain periods and spin states (Kaasalainen, 2004). Several groups have derived rotation periods or spin states from DR3, notably Ďurech & Hanuš (2023), Cellino et al. (2024) and MacLennan et al. (2026). Many asteroids with enough transits still have no published period.

Before such periods can be used, two things are needed. The first is a measured rate at which a period is wrong. The second is the same rate for asteroids that are *not* already in other catalogues. This paper provides both and makes the pipeline fast enough to rerun on Gaia DR4. A pilot on 600 asteroids was reported in Tropynin (2026a).

2. Data and geometry

All 20 bulk files of `sso_observation` (23.3 M CCD rows) were averaged per `transit_id`. This gives 3,069,170 transits of 156,789 numbered asteroids, of which 42,195 have at least 25 transits. Each transit carries the largest relative flux error of its CCDs, with a 0.03 mag floor. 353 asteroids fall below 25 transits after outlier clipping and are dropped.

Heliocentric positions come from JPL osculating elements (epoch 2026 June). These were integrated back to 2014–2017 with a fourth-order Runge–Kutta scheme, a one-day step, and the Sun and the giant planets as perturbers. Gaia's position is taken from the data, and light time is iterated. Against JPL Horizons on 19 asteroids, the reduced magnitudes agree to 0.0007 mag.

3. Model and search

The reduced magnitude $G-5\log_{10}(r\Delta)$ is fitted by weighted least squares. The model has:

- one constant per observing season;
- a term linear in phase angle;
- a second-order Fourier series in the phase $2\pi(ft - s\,\lambda_{\mathrm{PAB}}/360^\circ)$.

Here λ_{PAB} is the ecliptic longitude of the phase-angle bisector seen from Gaia (Harris et al., 1984), and $s = \pm 1$ is the sense of rotation. Both senses are searched, and the better fit is kept.

The search covers periods of 2–1000 h on a grid with step $0.5/T$, where T is the time span; the 30 deepest minima are then refined. A five times finer grid changed precision by less than a point. When the first harmonic exceeds 1.20 times the second, the doubled period is adopted.

The PAB correction was adopted because it is better on the tuning half. Over all solutions it raises the fraction of correct periods from 44.2% to 52.2%. At probability ≥ 0.8 it increases the number of periods from 928 to 1389 while keeping precision at 92%. The test half shows the same effect: 45.5% to 52.6%. With the correction, the median difference from the sidereal periods of Ďurech & Hanuš (2023) falls by a factor of four. Without it, that difference grows with period, as an uncorrected synodic drift would.

4. Calibration and null tests

The truth set is LCDB V4.0 (Warner et al., 2009, 2021) with quality $U \geq 2-$. It was released in September 2021, before Gaia DR3. A period counts as correct if it is within 3% of the LCDB value. The 9,613 known asteroids were split by a hash of their number into TUNE (4,779) and TEST (4,834).

On TUNE I fitted the doubling rule and a logistic model of the probability that a period is correct. The model uses seven features:

- the χ^2 gain;
- the separation from the next non-alias minimum;
- the number of transits;
- the amplitude;
- the reduced χ^2;
- the period;
- the harmonic ratio.

Table 1: Signal-free asteroids (out of 32,229) that reach a given probability in the null tests.

Threshold	White	Block
0.70	264	248
0.75	104	108
0.80	39	32
0.85	12	8

On TEST, the predicted probabilities match the observed fractions (Figure 3), but the top bin is slightly overconfident: 0.95 predicted against 0.89 observed.

The confidence threshold was not chosen on TEST. Two null tests use the real Gaia sampling but remove the rotation:

- *white*: residuals of the no-rotation model are permuted within seasons;
- *block*: one-day blocks of residuals are permuted, which keeps correlated systematics.

Noise periods match no catalogue, so all of them would land among the candidate new periods. The rule, fixed in advance, is that the projected noise must stay below 5% of that list. The probability grid step is 0.05.

Both null tests were run on all 32,229 solved asteroids without an LCDB period, so the projected noise count is simply the number that pass. At the TUNE threshold for 80% precision (0.384), 20.0% (white) and 18.1% (block) of these signal-free asteroids pass. Table 1 gives the counts at higher thresholds. Under the rule, the threshold is 0.80. There 39 (white) and 32 (block) asteroids pass, which is 4.5% and 3.7% of the 861 candidate new periods. The 95% Poisson upper limits are 51 and 43, that is 5.9% and 5.0%. These figures are conservative, because they treat every unknown asteroid as having no detectable rotation.

Tier A is probability ≥ 0.80 and period < 100 h. Tier B is probability between 0.384, the TUNE threshold for 80% precision, and 0.80. Tier B is not reliable for unknown asteroids, because 18–20% of signal-free asteroids reach it. Periods of 100–1000 h are flagged as slow candidates and never treated as confident.

5. Results

5.1. Periods

6,238 periods reach tier A. On TEST, 1303 of 1459 tier-A periods are correct (89.3%): 90%, 88% and 83% at 2–10, 10–30 and 30–100 h (Figure 2).

Precision depends on the quality of the reference:

- 95.4% (354/371) for $U \geq 3-$;
- 89.1% for $U = 2$ and 2+;
- 36% for $U = 2-$.

There are 156 disagreements:

- 58 differ from LCDB by exactly one cycle per day in lightcurve frequency, the daily alias of single-site photometry;
- 6 differ by a factor of two;
- 92 are other periods.

For 38 of the 156 asteroids an independent TESS period exists (Vavilov & Carry, 2025). It agrees with this work in 20 cases, with LCDB in 3, and with neither in 15.

The best analogue of the unknown population is the set of TEST asteroids absent from all six catalogues of Table 2. Tier A is correct for 185 of 205 of them (90%, 95% interval 86–94%).

The χ^2-curvature errors are too small. Against Ďurech & Hanuš (2023), the median $|\Delta P/P|$ is 3.9×10^{-5}. Adding $5.5 \times 10^{-5} P$ in quadrature gives a median $|\Delta P|/\sigma \approx 0.67$. This compares two methods applied to the same data and does not include aliasing.

5.2. Injection–recovery

I injected synthetic double-peaked lightcurves into the real Gaia sampling of 4,000 asteroids without a known period. The lightcurves had:

- periods of 2–100 h, log-uniform;
- amplitudes of 0.1–0.6 mag, scaled by 0.5–1 per season;
- a random spin sense, with the phase following λ_{PAB}.

Table 2: Agreement of tier-A periods with published catalogues (within 3%).

Catalogue	Data	n	Agree
Ďurech & Hanuš (2023)	Gaia DR3	3564	99.6%
Cellino et al. (2024)	Gaia DR3	2883	99.0%
MacLennan et al. (2026)	Gaia DR3	2676	99.0%
Vavilov & Carry (2025)	TESS	759	91.0%
Erasmus et al. (2020)	ATLAS	280	68.6%
McNeill et al. (2023)	TESS	863	55.0%

The Gaia DR3 rows test consistency, not accuracy. For the two lower rows, agreement rises with each catalogue's own confidence. It is 81% for Erasmus et al. entries with confidence $\geq 80\%$ ($n = 97$). It is 77% for McNeill et al. entries with $CV_{\text{exact}} \geq 0.9$ ($n = 419$). Factor-two relations make up $\leq 1.2\%$ in every row. For TEST asteroids that are also in LCDB, the two periods agree with LCDB as follows (this work vs. the catalogue): 373 vs. 359 of 411 (Vavilov & Carry, 2025), 111 vs. 116 of 146 (Erasmus et al., 2020), and 352 vs. 231 of 379 (McNeill et al., 2023).

The real noise lies between two models. In the pessimistic model the noise is the object's own permuted residuals; in the optimistic model it is Gaussian at the quoted errors.

Tier A is precise under both: 96.5% (109/113) and 99.3% (567/571) of its periods are correct. It contains no $P/2$ or $2P$ errors, and the spin sense is correct for 96–97% of the correct periods.

Completeness is low, 2.7% and 14%, compared with 10.5% of real unknown asteroids that reach tier A. It rises with amplitude and with the number of transits; no amplitude below 0.2 mag is recovered. Tier A is thus a high-precision, low-completeness selection. It should not be used for population statistics without a selection-function correction.

5.3. Spin senses

The sense s was compared with the hemisphere of published poles, excluding poles within 20° of the ecliptic and ambiguous pole pairs:

- 3473 of 3535 agree (98.2%) for Ďurech & Hanuš (2023), which uses the same data;
- 3802 of 3899 agree (97.5%) for DAMIT (Ďurech et al., 2010);

- 372 of 394 agree (94.4%) for DAMIT asteroids not in Ďurech & Hanuš (2023).

The errors are nearly symmetric. Against Ďurech & Hanuš (2023), 60 prograde asteroids are called retrograde and 44 retrograde asteroids are called prograde.

Tier A has 3,677 retrograde and 2,561 prograde rotators (59% retrograde). I do not interpret this excess here, because tier A is not an unbiased sample.

5.4. Yarkovsky segregation in families

The Yarkovsky drift moves prograde rotators outward and retrograde ones inward, so the two senses lie on opposite sides of a family's V-shape (Vokrouhlický et al., 2015). This has been shown with modelled poles for several families (Hanuš et al., 2013, 2018; Kryszczyńska, 2013).

I used AstDyS family membership and synthetic proper elements (Knežević & Milani, 2003; Milani et al., 2014), with the proper a of the namesake as each family's centre. A member is consistent with the effect if it is prograde and outside the centre, or retrograde and inside it. The right null hypothesis keeps each family's mix of senses and its positions and breaks only the link between them. I therefore permuted the senses within each family 20,000 times.

- **Overall.** 1,753 of 2,345 tier-A members (74.8%) are consistent, against 52.4±0.9% under the null hypothesis ($z = 24.7$; no permutation reaches the observed value). The 2×2 odds ratio is 9.1.

- **Distance from the centre.** The fraction rises from 58% for members near the centre to 82% for members whose $|a - a_c|\, D$ exceeds twice the family median, as expected for Yarkovsky drift.

- **Choice of centre.** The result is insensitive to it: 71.3% with the median a of all members.

- **Members without a previous pole.** The effect holds for the 912 members without a pole in Ďurech & Hanuš (2023): 73.6%.

- **Individual families.** The largest families show it clearly: (221) Eos 86% of 578, (24) Themis 81% of 184, (15) Eunomia 79% of 294, (158) Koronis 78% of 184 (Figure 1).

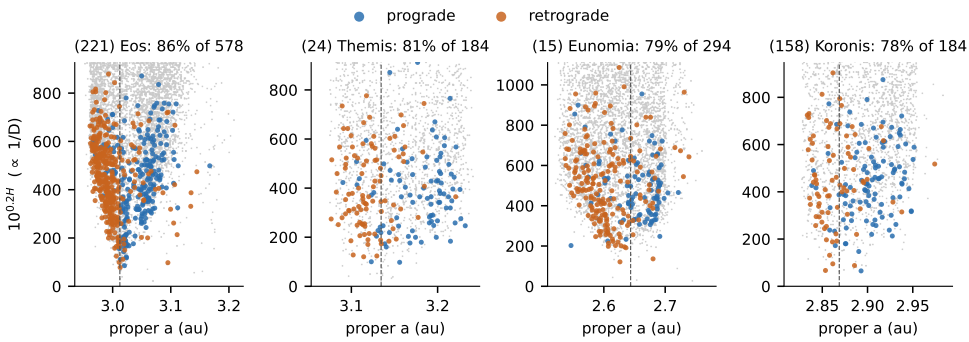


Figure 1: Four families in proper semimajor axis against $10^{0.2H} \propto 1/D$. Grey: all AstDyS members. Blue: prograde; orange: retrograde (tier A, this work). Dashed line: the namesake. Each title gives the fraction of members on the side the Yarkovsky effect predicts.

Random sense errors can only dilute the segregation. The observed 74.8% therefore implies that at least 75% of the senses are correct, a bound independent of any pole catalogue. Tier A favours high-amplitude rotators, so the fraction describes this subset, not complete families.

5.5. Disputed LCDB periods

I selected tier-A asteroids whose LCDB V4.0 period disagrees with this work and met one of two conditions:

- an independent survey supports this work and none supports LCDB (Vavilov & Carry, 2025; McNeill et al., 2023; Erasmus et al., 2020);
- at least two other Gaia DR3 analyses support this work.

This gives 117 asteroids. Of these, 32 are exact daily aliases and 14 differ by a factor of two.

In the 2023 LCDB revision distributed by the JPL Small-Body Database, 9 of the 117 have already been revised to the value found here. The other 108 are disputed. Of these:

- 39 are disputed by independent survey data and 69 by agreement of Gaia DR3 analyses only;

- by LCDB quality, 83 have $U = 2$, 9 have $U = 2+$, 5 have $U = 2-$, 3 have $U = 3-$ and 8 have $U = 3$.

The list is part of the catalogue.

5.6. Candidate new periods

942 tier-A asteroids have no period in LCDB or in the six catalogues. For 81 of them (8.6%), a rotation period exists in one of three other sources:

- DAMIT;
- about 180 VizieR tables with rotation periods (5.0 M rows), with units taken from the table metadata and non-rotation periods excluded;
- the Minor Planet Bulletin, volumes 50–53.

Where the values can be compared, every disagreement has an identified cause:

- partial-coverage PTF periods that are half of ours (Chang et al., 2016);
- K2 periods that differ by 3–6% (Sergeyev et al., 2025);
- fast rotators below our 2 h limit (Erasmus et al., 2019);
- two slow rotators.

The remaining 861 are candidate new periods: 667 at 2–10 h, 131 at 10–30 h and 63 at 30–100 h. The matched control gives an expected precision of about 90%. Noise is at most 32–39 periods (3.7–4.5%; 95% upper limit 5.9%). SsODNet, Behrend's Geneva page and ZTF period papers outside VizieR were not searched. Given the 8.6% found in the other sources, some further candidates are likely to be already published.

The pilot list of 42 candidates (Tropynin, 2026a) changes as follows:

- 35 agree with this work;
- 4 were half the true period;
- 3 (39932, 109412, 13686) are now ambiguous, with the extended search preferring slow solutions outside tier A.

6. Limitations and plans for Gaia DR4

- Periods below 2 h are not searched.
- The A_1/A_2 doubling rule is a heuristic.
- For doubled periods the PAB coefficient should be 2. A variant that also tries 2 changed TEST precision only from 89.3% to 89.7%.
- Tumblers fall outside a single-period model.
- Completeness is low below an amplitude of 0.3 mag, and nothing is recovered below 0.2 mag.

DR4 will cover about 66 months. For it, I plan to fit the sidereal period and the pole jointly on a coarse pole grid, with a model-based test of P against $2P$. Thresholds and null tests are fixed in advance, so that results can be released within a day.

CRediT authorship contribution statement

Oleg Tropynin: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of generative AI use

During the preparation of this work the author used Claude (Anthropic) to write and run analysis code and to draft and edit the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

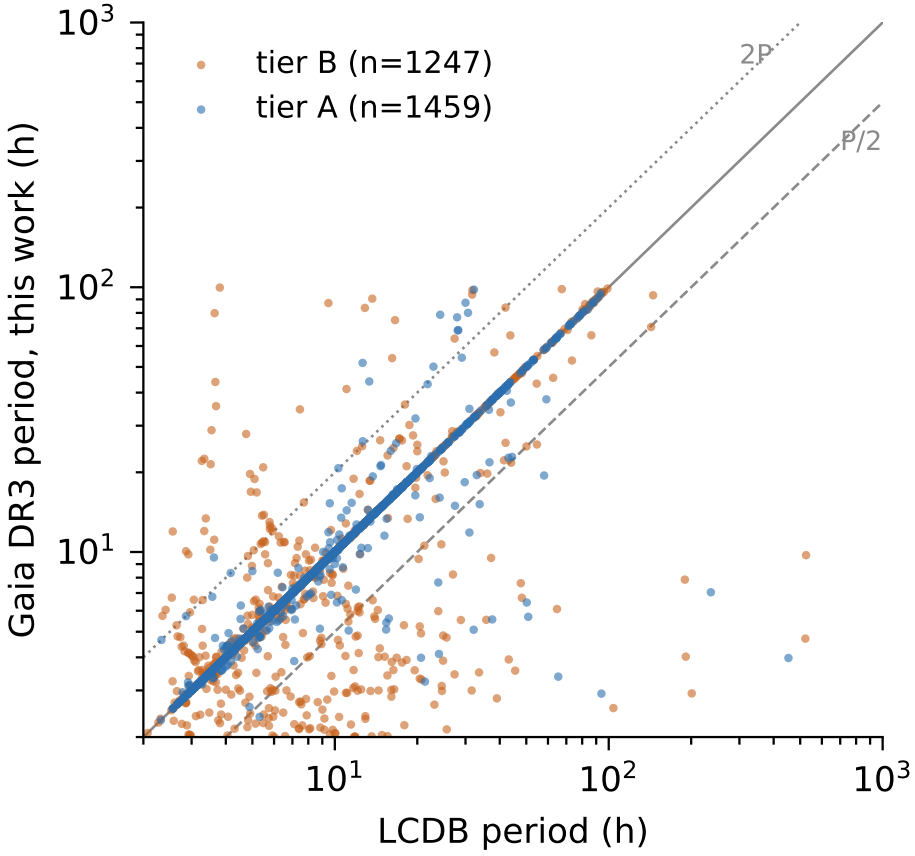


Figure 2: Period from this work against the LCDB period for TEST asteroids. Blue: tier A. Orange: tier B (probability 0.384–0.80). Lines mark P, $2P$ and $P/2$. The off-diagonal scatter includes daily aliases in the reference periods.

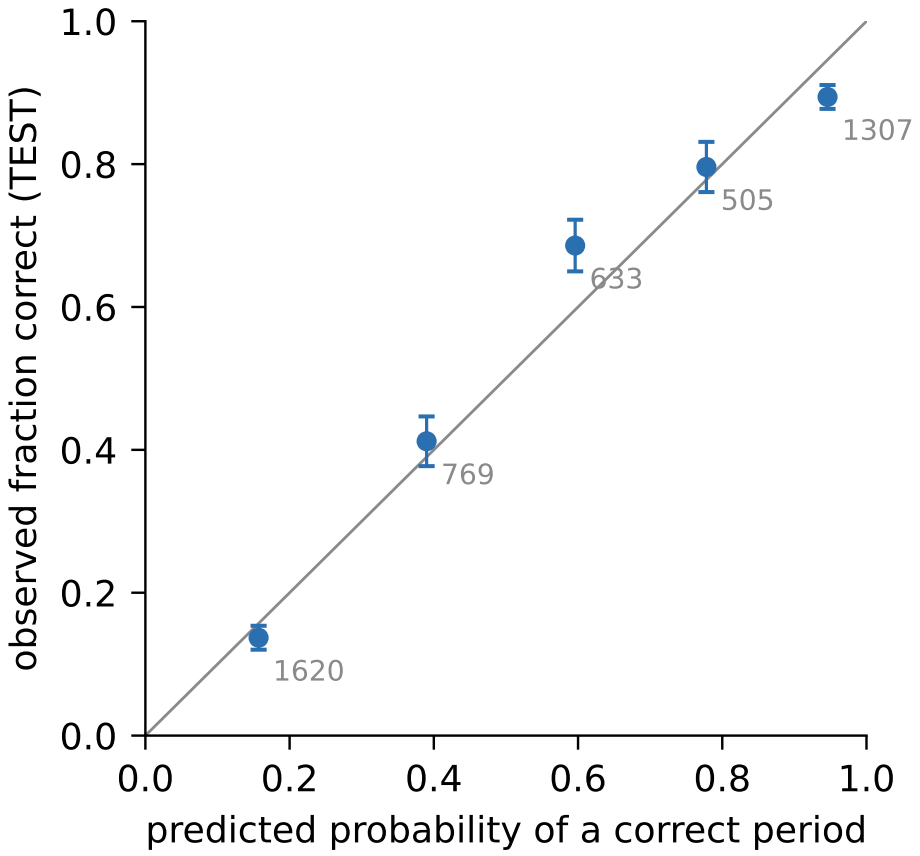


Figure 3: Reliability diagram on TEST for all solutions. Bars are 95% binomial intervals; numbers give the count per bin.

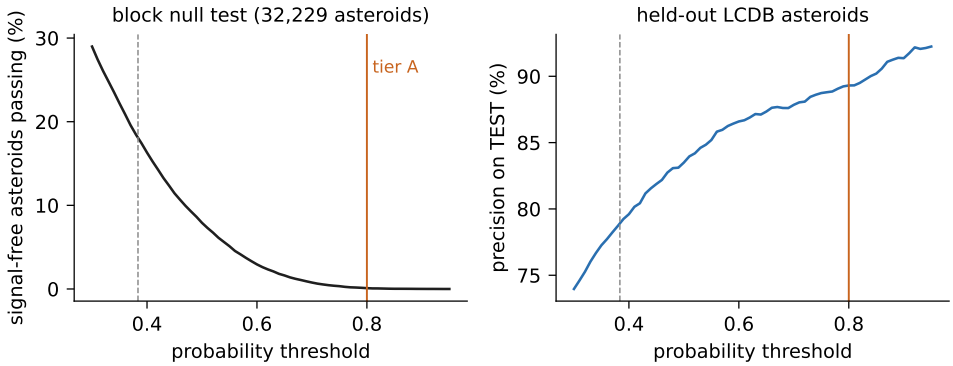


Figure 4: Left: fraction of signal-free asteroids (block null test) that pass, as a function of the probability threshold. Right: precision on TEST. Dashed line: the TUNE threshold for 80% precision. Solid line: the tier-A threshold.

Data availability

The catalogue (41,842 rows with periods, probabilities, tiers and spin senses), the list of 861 candidate new periods, the list of disputed LCDB periods, the validation tables and the full code are openly available on Zenodo, doi:10.5281/zenodo.23638281, and at `https://github.com/olegtropinin/gaia-dr3-rotation-catalog`.

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